



Spinach

CULTIVARS

- Red Kitten
- Space

INTENSIVE SPACING

- **Direct Sown:** 6 rows (5" - 12 cm) per bed spaced every 5" (12 cm) on the row
- - JANG 3X seeder: F-24, 14F-9B, no-brush "0"
- **Transplanted:** 5 rows (6"-15 cm), spaced every 6" (15cm) on the row

FERTILIZATION

- Spinach will do great in beds that were previously amended with compost and fertilized with a mix of alfalfa meal (2-0-2) and pelleted chicken manure (5-3-2).

IMPLANTATION

Because spinach doesn't do well in hot temperatures, we only grow it towards the tail end of the summer and all through fall and winter. We use a cut and come again strategy where we only harvest the large outer leaves of each plant over many weeks. To ensure a continuous supply, we stagger our planting to transition from field to winter greenhouse, planting enough beds to supply our weekly sales targets.

EQUIPMENT

- Rotary harrow
- 30" bed prep rake
- **Direct Seeding:** Individual Jang seeder + F24 roller, 5 tubes for the rake, scale, notepad and your seeds
- **Transplanting:** Popping tray and marking roller

TRANSPLANT ITINERARY

- Remove tarp and clean out large debris. Use the rake if needed. *
- Uniformly apply amendments and pass with the rotary harrow at a depth of 2" (5 cm). To make transplanting easier (seeing as the soil is loose after being harrowed), we only prepare the beds when we are about to transplant.
- Pass with the dibbler-roller to mark the planting holes and bring out the well-watered trays.
- Once the beds are ready, 2 people begin to drop seedlings while another person starts to plant. Use the popping tray to facilitate extraction. 3 people/ 100' bed is a good ratio for efficient workflow.
- Install irrigation and water until the soil is deeply moistened.

* If the previous crop is too well established, or if the bed is too messy, mow the crop as low as possible and harrow at a depth of 2" (5 cm). Add 1" (2.5cm) of compost evenly spread on the whole width of the bed and transplant into compost following a no-till strategy.

DIRECT SEEDING ITINERARY

- Remove tarp and clean out large debris. Use the rake if needed.*
- Uniformly apply amendment and pass with the rotary harrow at a depth of 1.5" (4 cm). Use the rake to remove debris if needed. Otherwise, the bed is ready to be seeded.
- Seed with the Jang, writing down the total amount of seed dropped onto the first bed and comparing with the target. Adjust calibration if required.
- Install irrigation and water until the soil is deeply moistened. Keep moist for the next 7-10 days until all seeds have germinated
- Start some extra seedlings in the greenhouse to fill in the holes on the beds. These should be started the same week as when the direct seeding is done.
- Spinach seed is short-lived. For optimal germination, start with new seed each year.

*If the previous crop is too well established, or if the bed is too messy, mow the previous crop as low as possible, harrow the beds at a depth of 2" (5 cm) and rake it clean. Add 1" of compost on the bed and level well using the back of the rake to get the same thickness across the whole width of the bed. Seed directly unto compost following a no-till strategy

NOTES FOR OVERWINTERING SPINACH

- Plant growth almost stops when daylight is less than 10 hours per day. Research when this happens in your region and schedule planting in order to have 4-5 true leaves on your plants before this point.
- In our northern area, plants need to be established by early November. If direct seeding, a September seeding date is required. However, germination can be poor in tunnels with warm soil temperatures (>70F/21C). Transplanting seedlings instead of direct seeding can help mitigate this risk, and ensures a good stand for winter production.
- Highest total yields (fall through spring) will be obtained by the earliest fall planting dates, primarily due to higher fall production. Spring-only yields (January - April) are less affected by fall planting date.

WEEDING

- Spinach is cultivated using the flex tine weeder 10 days after transplant (14 days when direct sowed) and again 7 days later..

PLANT PROTECTION

- Downy mildew is a common condition favoured by the cool, moist conditions in the winter greenhouses. Relying on mildew-resistant varieties is your best bet. Once infected, there is little that can be done.

HARVEST

- Spinach is harvested in the fields in a kneeling position.
- Harvest by removing only the outer leaves, and not the center leaves. This will allow the plant to keep producing.
- Harvest the leaves that are 3 -4" long (8-10cm). They should be dark green and free from any white mold or yellowing.

- Make sure to pick using both hands, carrying as many leaves as possible before dropping in the harvest containers.
- 1 person should harvest 80 lbs per hour.

CONDITIONING

- Spinach is washed by quickly dunking leaves in a Bubbler filled with cold water.
- Make sure to sanitize all the surface areas that come in contact with the spinach.
- Do not leave greens for longer than 1 minute in water. Leaves tend to get waterlogged after a few minutes, which make them deteriorate faster.
- Spin dry the spinach and pack in sealed plastic bags or storage totes
- 1 person should clean at least 100 lbs per hour.

STORING

- Spinach is stored in cold room kept at 35°F (2°C) and should not stay longer than 3 days before being sold.